

MINDCRAFT

General Learning Science Digest

General cognitive science and educational psychology findings, useful for anyone building a learning app, stripped of MindCraft's own product framing.

Source: 34 chapters of the Research Constitution, ~3,031 words in this digest.

General Learning Science, for Anyone Building a Learning App

This is not MindCraft's product strategy. It is the underlying cognitive-science and educational-psychology research sitting inside MindCraft's internal research corpus (agent_work/research/, 160 chapters), stripped of MindCraft's own jargon, product names, and internal codes. Every bullet traces to a real, citable finding from a specific chapter, marked in parentheses. Meta-analyses and large replicated findings are prioritized over single small studies, and single-study findings are marked as such.

How memory actually works

- Spacing practice out over time beats massing it into one sitting, and the ideal gap between sessions gets longer the longer the material needs to be retained. Well established at large scale (Cepeda et al 2006, 317 experiments; math-specific replication, Murray, Horner and Goebel 2025, 27 studies, modest effect $g=0.28$). (Ch 29, 78)
- Actively recalling an answer from memory beats rereading or passively restudying it (the testing effect), one of the most replicated findings in the field (Roediger and Karpicke 2006). (Ch 29)
- Forgetting follows a decelerating, somewhat irregular curve, not a smooth exponential drop, a genuine replication of Ebbinghaus's original 1880s work (Murre and Dros 2015; Wixted 2004 review). (Ch 76)
- People believe cramming works better than spacing even when spacing demonstrably wins for the large majority of them, a real, replicated metacognitive illusion (Kornell 2009: spacing beat massing for about 90% of participants, yet 72% believed massing worked better). Same-session confidence is a poor proxy for later retention, since judgments of learning made right after study ignore how much the coming delay will hurt recall (Koriat and Bjork's foresight bias research). (Ch 78, 76)
- Personalizing which item to review, based on what a learner is actually closest to forgetting, beat both cramming and a generic fixed spaced schedule in a real semester-long classroom study (Lindsey et al 2014, about 16.5 points better than massed practice on a month-later exam). Engagement

or predicted-recall accuracy improving is not the same evidence as learning improving (a caution drawn directly from Duolingo's own published algorithm research). (Ch 77)

- Mixing different problem types together (interleaving) produces substantially better delayed-test performance than grouping same-type problems together (blocking), including in a preregistered, classroom-scale randomized trial (Rohrer et al 2020, 54 classrooms, $d=0.83$ on a one-month test). This depends on the material: it helps most when categories are easily confused, and can actually hurt for some material like word lists (Brunmair and Richter 2019 meta-analysis, 59 studies). (Ch 39, 114)

Difficulty and struggle

- A "desirable difficulty" (spacing, interleaving, retrieval practice) only helps if the learner already has enough prior knowledge to respond to it successfully. The identical difficulty is simply harmful for someone who isn't ready for it yet (Bjork and Bjork's foundational framework). (Ch 41, 114, 117)

- For genuine novices, studying complete worked examples beats unguided problem-solving, because unaided search consumes the working memory needed to actually build understanding (Sweller and Cooper 1985, foundational cognitive load theory). This reverses once a learner gains real competence, where continuing to hand-hold with examples becomes redundant and can hurt (the expertise reversal effect, Kalyuga et al 2001, 2003). (Ch 26, 88, 100)

- Fading support gradually, removing the final steps of a worked example first rather than jumping straight from a full example to an unaided problem, produces better transfer, and adapting the fade schedule to actual demonstrated performance beats a fixed fade schedule (Renkl et al 2002; Salden et al 2010, intelligent tutoring system study). (Ch 88, 100, 117)

- Letting students attempt a novel problem before any instruction, then explicitly comparing their attempts to the correct method, can beat teaching first, but only under real conditions: older learners, conceptually rich material, and a genuine consolidation step afterward. It reverses for young children and for generic (non-domain) skills (Sinha and Kapur 2021 meta-analysis of 53 studies, $g=0.36$ overall, moderated heavily by fidelity

and age). (Ch 94)

- Adding more explanatory content is not automatically better. Current cognitive load theory treats "germane load" as a redistribution of existing mental effort, not a third type you can just add more of, and detailed provided explanations add close to nothing once a learner is already prompted to self-explain (Wittwer and Renkl 2010 meta-analysis, 21 studies, small overall effect $d=0.16$, near zero once self-explanation is already present). Sometimes withholding an explanation and requiring the learner to generate it produces better conceptual learning than providing it (Richey and Nokes-Malach 2013). (Ch 92)

- More assistance is not automatically safer. Over-assisting can undercut the very constructive effort that produces learning, a genuine, still-debated design tension in the tutoring-systems literature, not a settled formula (Koedinger and Aleven 2007's "assistance dilemma"). (Ch 92)

- For learners showing real anxiety on a topic specifically, providing a full worked example at that moment measurably blunts anxiety's damage to learning and reduces mind-wandering, a legitimate support, not coddling (Mesghina et al 2023, one study, 280 fifth graders). Readiness to remove that support is better gated on demonstrated success than on a fixed anxiety score. (Ch 117)

Motivation and identity

- Whether someone keeps engaging with a subject is driven by three independent, separately measurable things: expectancy (do I believe I can succeed), value (is this interesting or important to me), and cost (effort, emotional cost, what else I'm giving up), not one general "motivation" level (Eccles-Parsons's expectancy-value model, decades of supporting research). A short prompt asking someone to connect material to their own life raises interest and performance the most for people who don't expect to do well (Hulleman and Harackiewicz 2009, a real randomized field experiment). (Ch 37)

- Framing progress around improvement against your own past performance (mastery goals) produces better strategies and more resilience to failure than framing it around outperforming others or proving you're smart

(performance goals), a large, well-established body of evidence (Ames and Archer 1988; Dweck and Leggett 1988; Hulleman et al 2010 meta-analysis of 243 studies, about 91,000 people). Goals framed around avoiding looking incompetent are broadly harmful; the actual reward and evaluation structure of an environment shapes this more reliably than messaging alone (Ames 1992; a real longitudinal classroom study, Lueftenegger et al 2014, 1,680 students). (Ch 38)

- Autonomy, competence, and relatedness are the three psychological needs that sustain real intrinsic motivation (Self-Determination Theory, Ryan and Deci 2000, one of the most cited frameworks in psychology). Rewarding mere completion of a task (not real performance) reduces later interest in that task once the reward is gone, an effect not seen for unexpected rewards or informational, non-controlling praise (Deci, Koestner and Ryan 1999 meta-analysis, 128 studies). Choice increases motivation and effort, but attaching an immediate reward right after the choice cancels much of that benefit (Patall, Cooper and Robinson 2008 meta-analysis, 41 studies). (Ch 44)

- Habits are triggered by a stable context cue, not by ongoing willpower or motivation, so changing the cue or environment is the real lever, not "trying harder" (Wood and Runger 2016 review). Specific "when X happens, I will do Y" plans reliably increase follow-through, one of the more robust findings in behavior change (Gollwitzer and Sheeran 2006 meta-analysis, 94 tests, d is about 0.65). Missing a single day does not meaningfully derail habit formation, and full automaticity takes far longer than the popular "21 days" claim, a median of about 66 days with huge individual variation (Lally et al 2010, real-world diary study). (Ch 43)

- Resilience under real adversity typically comes from ordinary protective systems staying intact (attachment, self-regulation, social support), not from a rare personality trait (Masten 2001's widely cited "ordinary magic" framing). "Grit" specifically has weak evidence as a distinct, trainable trait: it overlaps heavily with the existing trait of conscientiousness, predicts achievement only modestly, and grit-building interventions show weak effects (Crede, Tynan and Harms 2017 meta-analysis, 584 effects across about 66,800 people). (Ch 45)

- Curiosity is triggered by a perceived, specific information gap and is intense but short-lived, a matter of situational design more than a personality trait (Loewenstein 1994's foundational framing). Committing to an explicit guess before seeing an answer increases curiosity and improves memory for the answer, especially when it's surprising, and asking "were you surprised" after the fact does not reproduce this effect (Brod and Breitwieser 2019). (Ch 107)

Feedback and metacognition

- Feedback has a real positive effect on performance on average, but it is inconsistent: more than a third of feedback interventions studied actually made performance worse, especially feedback that draws attention to the person or their ego rather than the task itself (Kluger and DeNisi 1996, one of the most cited meta-analyses in the feedback literature, 607 effect sizes). There is no universal rule that immediate or delayed feedback is always better, it is genuinely context-dependent, and a recent meta-analysis found the average difference between the two statistically indistinguishable from zero (Kulik and Kulik 1988; a 2024 meta-analysis of 116 interventions). Feedback aimed at the task or strategy ("try this differently") is more useful than feedback aimed at the person ("great job," comparative praise) (Hattie and Timperley 2007). (Ch 79, 112)

- People are frequently overconfident, especially when their actual accuracy is near chance, a foundational and widely replicated pattern that also shows up in school-age math learners specifically (Lichtenstein and Fischhoff 1977; Garcia et al 2016). Confidently held wrong answers are corrected more durably by feedback than uncertain wrong guesses, the hypercorrection effect, now replicated in real, large-scale online math data (Butterfield and Metcalfe 2001; Foster et al 2022). Just asking someone to rate their confidence does not, by itself, improve performance, a real null result from a study across four schools, it only helps when paired with feedback or a decision that changes what happens next (Foster 2021/2022). Prompting someone to generate reasons their answer might be wrong, not just reasons it's right, meaningfully improves the accuracy of their confidence (Koriat, Lichtenstein and Fischhoff 1980). (Ch 49, 50)

- Formative assessment's real, validated definition is a process, evidence gathered, interpreted, and used to change the next step, not any specific artifact like a quiz (Black and Wiliam 1998, a landmark review of about 250 sources). The widely quoted "0.4 to 0.7 effect size" for formative assessment is not well supported by rigorous reanalysis; the real number is closer to 0.20 to 0.25 and still contested (Kingston and Nash 2011). Feedback only functions as formative if the learner understands the target standard, compares their own work to it, and takes an actual action to close the gap; feedback that leads to no action is close to worthless (Sadler 1989's foundational framing). (Ch 111)

- Students often say feedback would be useful but admit they would not actually use it. The real barriers run through noticing it, understanding it, feeling able to act on it, and being willing to put in the effort, which means low uptake is usually a design failure, not a character flaw in the student (Winstone et al 2017, a systematic review of 195 outputs, plus a matching focus-group study). (Ch 112)

- Self-regulated learning runs as a repeating cycle: planning and goal-setting before a task, monitoring and control during it, and reflection afterward, where that reflection feeds the next attempt's plan (Zimmerman's foundational model). Attributing a failure to a fixable cause, like the wrong strategy, sustains motivation and produces better adaptive change than attributing it to fixed ability (Zimmerman 2002; Cleary, Zimmerman and Keating 2006). (Ch 101)

Social and affective context

- Math anxiety correlates with worse performance, and one credible mechanism is that anxiety-related worry competes for the same working memory the task itself needs, not simply that anxious people have lower ability (a meta-analytic correlation around $r=-0.17$; Ashcraft and Kirk 2001 found reduced working-memory span specifically under anxiety). Sleep deprivation independently and measurably harms both new learning and retention of recent learning, especially in teenagers: five nights at five hours of sleep caused 26 to 65% more forgetting than well-rested peers depending on the delay tested (Cousins, Wong and Chee 2019). (Ch 24, 47)

- Stereotype threat and belonging interventions are real, well-studied effects, but they are genuinely heterogeneous and often shrink once scaled beyond the original controlled study, not a guaranteed, portable fix (Steele and Aronson's tradition; Walton and Cohen's belonging trials). (Ch 24)

- A learning environment's "error climate," whether mistakes are consistently treated as safe and useful, is a real, measurable construct that predicts how adaptively people respond to their own errors, beyond general motivation, and it decays over time unless someone actively maintains it (a validated scale across 1,116 students and 56 classrooms; a 2-year longitudinal follow-up of 1,641 students). How an instructor specifically responds to one person's error, staying to diagnose it versus redirecting away or giving a bare "incorrect", is itself a causal lever on how safe errors feel, demonstrated in a controlled experiment, not just observed correlation. (Ch 103)

- Structured collaborative learning shows real, moderate average benefits for STEM and math specifically, but unstructured "work together" time does not reliably produce this, structure is the active ingredient (multiple meta-analyses converging around 0.4 to 0.6 effect sizes; a math-specific meta-analysis at $d=0.59$). A group's shared performance score is never valid evidence of what any individual actually learned, since students who solved it, students who learned through the process, and students who were simply told the answer can all produce the same group score (Webb 1993). (Ch 113)

- Low-stakes retrieval practice tends, on average, to reduce test anxiety rather than increase it, the opposite of the intuitive fear that testing makes people more nervous (a meta-analysis of 24 studies, g is about -0.52). High-stakes framing of a practice session, however, can cancel out the normal memory benefit of retrieval practice even when in-the-moment performance looks unaffected (Hinze and Rapp 2014). (Ch 41, 114)

Practical instructional design

- Prompting learners to explain the reasoning behind a step, not just showing them the worked solution, improves understanding and transfer, and structured formats (choose the right principle from options) have more reliable supporting evidence than open-ended "explain your answer" boxes

(Chi et al's classic studies; a ZDM meta-analysis by Rittle-Johnson, Loehr and Durkin 2017). Asking someone to justify an answer before they know if it's correct can backfire by entrenching wrong reasoning, it works better to reveal correctness first, then ask for the explanation. (Ch 40)

- There is a real, well-studied difference between instrumental help-seeking (getting just enough to proceed) and executive help-seeking (getting the task done for you), and racing straight to hint-level answers correlates with worse learning across multiple tutoring-system studies. Help avoidance is just as real a failure mode, and is more common precisely among the lower-prior-knowledge students who need help most, so a system should not assume people will ask when they need it. (Ch 89, 104)

- Real scaffolding is contingent (support increases when someone struggles and decreases when they succeed), fades over time, and transfers responsibility to the learner. A fixed level of help that never changes is not really scaffolding in the technical sense (Wood, Bruner and Ross's foundational 1976 paper). Revealing a full answer is not inherently harmful, the harm comes from receiving it with no engagement, pairing a full reveal with a required self-explanation step can preserve most of its instructional value. (Ch 91, 95)

- Generative activities, summarizing in your own words, self-testing, self-explaining, teaching someone else, physically enacting an idea, reliably outperform passive study across a wide research synthesis, with moderate to large effect sizes depending on the strategy (Fiorella and Mayer's synthesis of eight strategies). Merely expecting to teach later does not produce this benefit, actually producing the explanation is what matters, and peer tutors themselves default to reciting rather than building understanding unless specifically prompted with inference-requiring questions. Free-form "draw whatever helps" underperforms a well-made provided illustration unless the drawing task is genuinely structured (partial templates, guided comparison). (Ch 102, 106, 109)

- Analogy works through matching the relational structure between two cases, not surface similarity, and putting both cases side by side with an explicit "what maps to what" prompt produces better transfer than studying them separately (Gentner's foundational theory). Having a structurally

similar example available does not guarantee someone will spontaneously notice and use it, an explicit cue is usually needed, a classic and frequently replicated finding (Gick and Holyoak 1980). (Ch 108)

- Predicting an outcome before watching a demonstration produces substantially better understanding than passively watching it play out (Crouch, Fagen, Callan and Mazur, a well-replicated classroom physics finding). A correct, strategy-specific hand gesture during instruction can improve learning beyond speech alone, and abstract or representational gesture supports transfer to new problems better than literal hands-on object manipulation, though forcing gesture on someone who doesn't spontaneously gesture can actually hurt them relative to an alternative scaffold, so it should not be mandatory for everyone. (Ch 107, 105)

- Unrestricted generative-AI assistance during practice raised in-session scores but measurably harmed later unassisted exam performance; adding pedagogical guardrails prevented that harm but still failed to beat a no-AI control on actual learning, a real, well-powered randomized study with about 1,000 high school students (Bastani et al 2025, published in PNAS). Practice accuracy is only meaningful evidence of learning if genuine cognitive effort is actually happening: when correct answers become trivially available to copy, homework scores can fully decouple from exam performance in the same course. (Ch 104)

- Human and AI tutoring's real average benefit is far smaller than the popularized "two sigma" claim; modern meta-analyses converge on roughly 0.37 to 0.79 standard deviations depending on method, and feedback at the step level beats feedback only on the final answer. Similarly, "deliberate practice" (coach-designed, just past current ability, with real feedback) explains far less of expert performance than popularly believed, about 4% of variance in one large educational meta-analysis, and raw time spent studying does not predict achievement once focus and goal-directedness are accounted for. (Ch 26, 51)

*Source: 34 chapters read directly from agent_work/research/chapters/ across five parallel extraction passes. Every finding above traces to a real citation

in a specific chapter; go to that chapter file directly for the full citation, effect size, and any caveats the original researcher noted.*